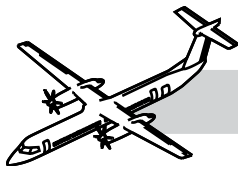


CHAPTER 80 ENGINE STARTING

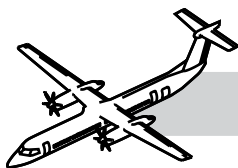
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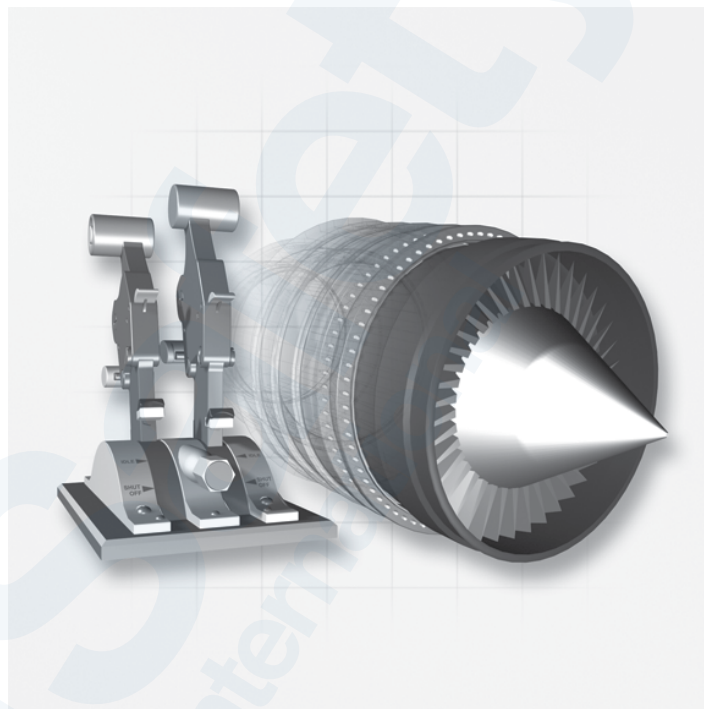
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CHAPTER 80

ENGINE STARTING

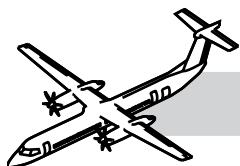


80-00-00 INTRODUCTION

The engine start system starts the engines on the ground and in the air.

SYSTEM DESCRIPTION

The engine control system is powered up when the appropriate airframe 28 VDC Essential Bus is selected ON.



COMPONENT DESCRIPTION

ENGINE START CONTROL PANEL

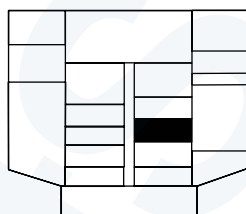
Refer to Figure 80-1. Engine Start Control Panel.

This panel, on the Overhead console, provides controls and indications for engine starting.

With the ignition switches selected to the NORM position, an engine start can be initiated by selecting the engine and activating the starter by pressing the engine START switch and moving the Condition Lever from the FUEL OFF, to START FEATHER position.

The engine start sequence is:

- Select NORM 1 and/or 2
- Set SELECT switch 1 or 2 (SELECT segment of START push light comes on)
- Push START (START segment of push light comes on)
- At 50% engine speed, the START light goes off, and the SELECT switch is released to the center position.



OVERHEAD CONSOLE

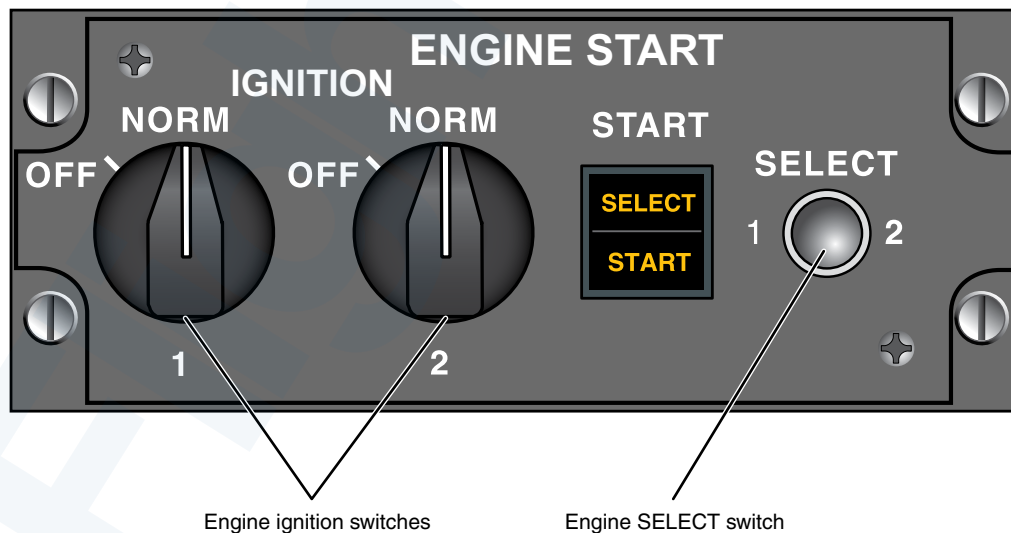
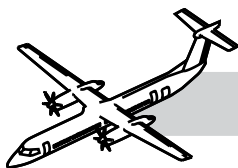


Figure 80-1. Engine Start Control Panel



Automatic Engine Starts on the Ground

When the initial engine start is successful, the FADEC controls the starting sequence as follows:

- When the starter has increased the generator speed (N_h) to 8%, the FADEC commands ignition On and schedules fuel flow as a function of N_h , ambient conditions, and Main Oil Temperature (MOT)
- Only one of the two ignitors is turned on. If the engine does not light within 8 seconds of fuel flow on, the FADEC turns on both ignitors
- Starts a FAULT COUNT for the faulty ignitor. Light off of engine is defined as an increase of 20°C ITT
- During run-up to idle, the FADEC switches channels to test the opposite channel
- When N_h is greater than 50%, exciter (and ignitors) are automatically de-energized
- The FADEC controls engine run-up to the commanded N_h idle speed of 64% N_h .

During ground starts, the FADEC actively limits the ITT.

The FADEC reduces the fuel flow below the standard start schedule (if necessary) to prevent an overtemperature.

The FADEC automatically aborts the start, and shuts down the engine if any one of these conditions occur:

- The engine does not light up within 16 seconds of fuel flow on
- ITT limit of 920°C is exceeded
- N_h does not reach 50% within 70 seconds (i.e., hung or slow start).

Ignition Control During Starts

The dual channel ignition system is powered electrically by the aircraft essential buses and activated by the FADEC.

The ignition system is configured such that both ignitors can be commanded by each FADEC channel, while maintaining electrical isolation between the FADEC channels and the aircraft essential buses.

The ignition selection from the flight compartment is sent to the FADEC on an ARINC 429 bus through the ECIU.

The FADEC ignition system has two flight compartment selectable modes of operation:

- 1-OFF: The FADEC disables ignition regardless of ground or flight status
- 2-NORM: The FADEC activates ignition during engine starts (ground or flight starts). The FADEC commands both ignitors On during flameout and surge accommodation.

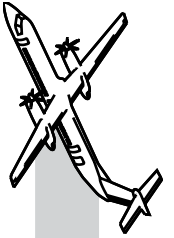
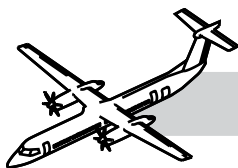


Figure 80-2. Engine No.1 Selected on Start Select Switch



ENGINE START

NOTES

Engine No.1 Selected on Start Select Switch

Refer to Figure 80-2. Engine No.1 Selected on Start Select Switch.

28 VDC is always available at Engine Start Panel SELECT switch Pins 3 and 6 as well as to time delay relay 3-K3 when the Essential bus is powered.

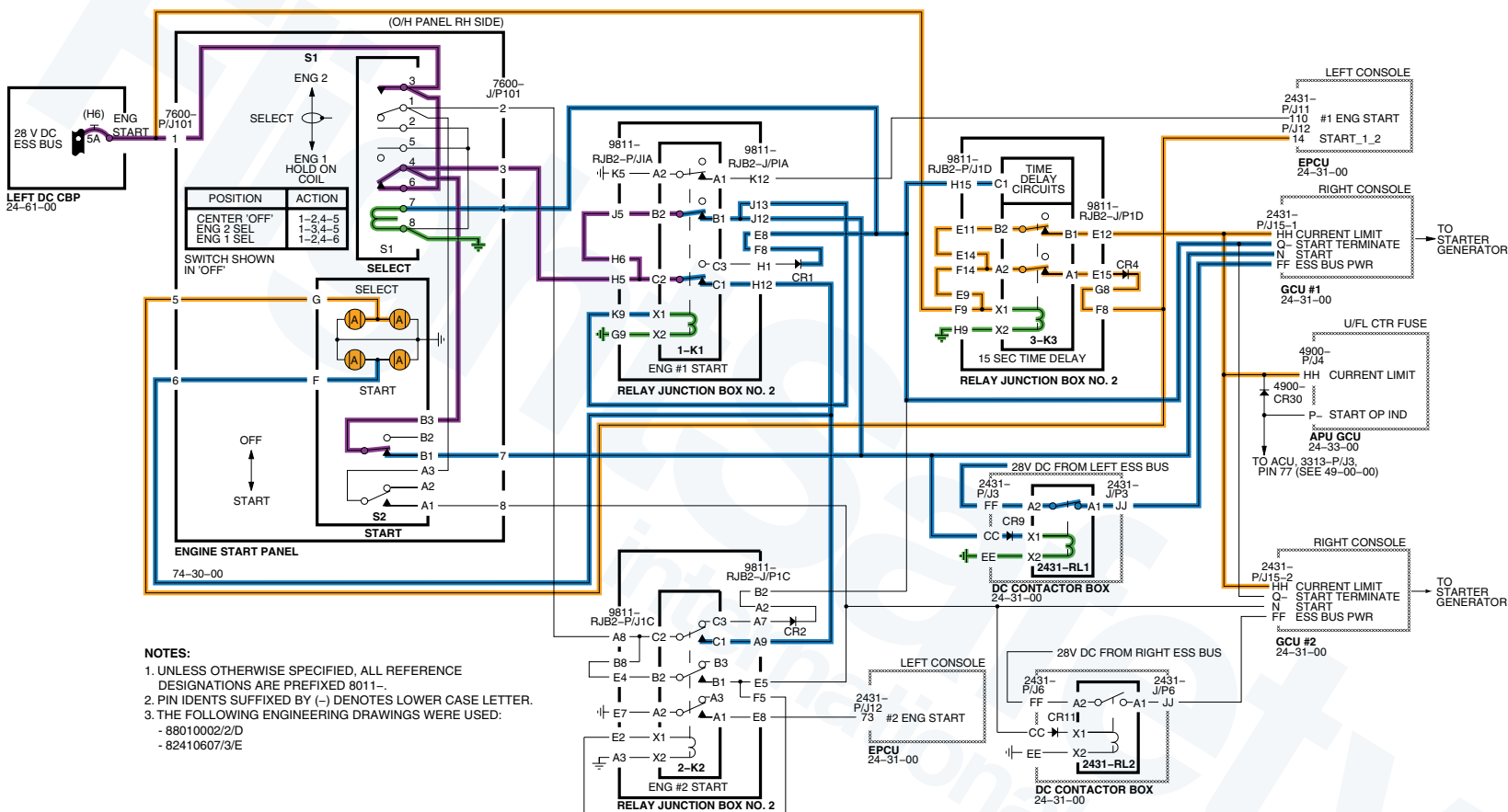
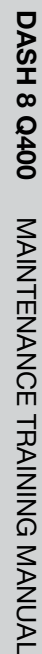
When the Engine Start Panel SELECT switch is set to Engine No.1:

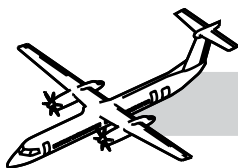
Select switch contacts 4 to 6 close, 28 VDC from the Essential Bus through the Start Select Switch is applied to the:

1. The START switch, arming the start circuit
2. The Holding coil of the Select Switch, holding the switch in the selected position
3. Contact B2 and C2 of de-energized ENG No.1 Start relay 1-K1, and through the de-energized contact of C2-C3 of 1-K1 to:
 - A. Control pin (C1) of the 15 second Time delay relay 3-K3
 - B. Start terminate input of the L/H and R/H GCU.

With 28VDC applied to the control pin C1, relay 3-K3 energizes providing 28 VDC to the:

1. Engine Start Panel S2 - SELECT light is ON
2. EPCU Start 1-2 input, EPCU enters the start monitoring mode
3. Current limiting input of the GCUs, arming the current limiting function of the GCU.





Start Switch Pushed (Momentary Action)

NOTES

Refer to Figure 80-3. Start Switch Pushed (Momentary Action).

1. Energizes Start Relay 1-K1 to initiate the start process. Energizing 1-K1 will:
 - A. Turn the start light ON
 - B. Supply power to the start Pin of the L/H GCU
 - C. Provide a holding circuit for relay 1-K1
 - D. Provide a ground signal to the EPCU for No.1 Engine Start pin of EPCU for monitoring purposes.
2. Energizes RL1 to supply Essential Bus power to the GCU
3. Supplies power to the START pin of the GCU to initiate the START mode
4. GCU controls the START TERMINATE sequence, providing 28VDC to C1 of 3-K3, and to the coil of S1 to hold the Start Select switch in the selected position.

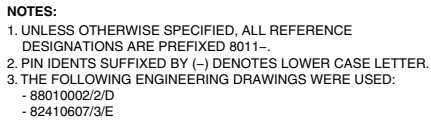
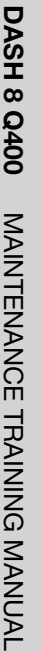
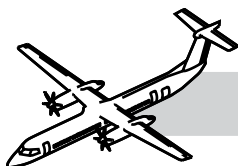


Figure 80-4. Start Switch Released



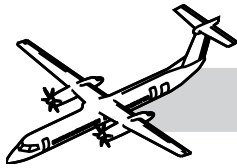
Start Switch Released

NOTES

Figure 80-4. Start Switch Released.

1. Holding circuit keeps 1-K1 energized
2. 3-K3 and holding circuit in S1 remain energized
3. When Nh reaches approx 50% the GCU removes the power output from the START TERMINATE pin
4. S1 holding circuit is released the SELECT switch returns to OFF
 - A. Left Essential bus power through RL1 is removed from the GCU
 - B. 1-K1 de-energizes immediately
 - C. START light goes off
 - D. 3-K3 remains energized for 15 seconds
5. After 15 seconds the GCU Current Limiting signal is removed
 - A. The Select light goes off
6. EPCU Start Monitoring Signal (No.1 Engine Start) removed
7. Start is completed
8. Nh rises to 64%.

When engine No.2 is selected the sequence is similar.



Flight Mode and Ground Mode starting are determined by:

- Flight Mode is set if the Calibrated Air Speed (CAS) is greater than 75 kt (139 km/h)
- In fault conditions when CAS is not available, the WOW Discrete from the Propeller Electronic Control (PEC) determines Ground/Flight Mode Status.

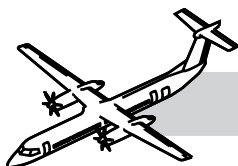
Automatic Starts in Flight

In-flight starts are similar to ground starts except for:

- The two ignitors are commanded ON
- Automatic ignition in flight is initiated when Nh drops to 60%.
- If engine relight is successful ignition is automatically stopped.
- If auto relight is unsuccessful ignition is terminated when Nh falls to 30%
- The auto-abort features are disabled
- FADEC does not actively limit ITT
- No FADEC channel transfers during the start.

CAUTION

DO NOT OPERATE THE ENGINE ON THE GROUND IN CONTINUOUS CROSSWINDS OF MORE THAN 50 KNOTS OR IN GUSTING CROSSWINDS OR MORE THAN 55 KNOTS. DO NOT OPERATE THE ENGINE ON THE GROUND AT ENGINE POWER ABOVE 460 SHP IN CROSSWINDS OF MORE THAN 45 KNOTS. IF YOU DO NOT OBEY THESE LIMITATIONS, YOU MUST REMOVE THE PROPELLER FROM SERVICE WITHIN THE NEXT 10 FLYING HOURS. DO NOT USE MORE THAN 14% ENGINE TORQUE WITH UNDERSPEED GOVERNING AT 660 RPM. THIS WILL MAKE SURE THAT THE ENGINE POWER IS NOT MORE THAN 460 SHP.



80-00-00 MAINTENANCE PRACTICES

Refer to the Bombardier AMM PSM 1-84-2 for details on these maintenance procedures:

- FIM 80-00-00-810-801: SELECT, START, ENGINE START panel with the SELECT toggle switch set to the 1 position (Caution) - Fault Isolation.
- FIM 80-00-00-810-802: SELECT, START, ENGINE START panel with the SELECT toggle switch set to the 2 position (Caution) - Fault Isolation.
- FIM 80-00-00-810-803: SELECT Light on the ENGINE START Control Panel Stays ON - Fault Isolation.